Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_08

1. Summary of Key Deviations

The analysis of Run_08 reveals a single, primary anomaly: a consistent and moderate loss of pressure in the combustion chamber. All other deviations appear to be minor or negligible.

- **p_Combustion (Moderate):** The only significant anomaly is a moderate **negative offset** on the combustion pressure, which reads approximately **-10%** lower than healthy across all active states (s2–s7).
- **Flow_Bypass & Flow_Pump (Minor):** Both of these flow signals show a very slight **positive offset** (around +4% to +6%). This is likely the control system's minor, and largely ineffective, attempt to compensate for the pressure loss.
- **All Other Signals (Nominal):** All other signals, including all four `Flow_Engine` sensors and all other pressure sensors, show no significant deviations.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Combustion Chamber / Injector Nozzle(s)**	**Minor Leak**	This is the most compelling hypothesis. A small but persistent leak in the combustion chamber itself or at one or more of the injector nozzles would cause a direct loss of pressure, perfectly matching the **-10% offset on p_Combustion**. The control system, detecting this pressure drop, would slightly increase pump flow to compensate, explaining the minor positive offsets on `Flow_Pump` and `Flow_Bypass`.

2	**p_Combustion Sensor**	**Calibration Shift (Negative Offset)**	It is possible that the `p_Combustion` sensor has a minor negative calibration error, making it *appear* as if pressure is low. Under this assumption, the slightly increased flows could be the control system reacting to this (erroneously) low-pressure reading. However, a real physical leak is a more direct explanation for the combined symptoms.
3	**Fuel Metering Unit (FMU)**	**Minor Under-delivery**	If the FMU were delivering slightly less fuel than commanded, it could lead to lower combustion pressure. However, in this case, we would expect to see a negative offset on the `Flow_Engine` sensors, which are all nominal. This makes an FMU fault less likely.
4	**Bypass Valve**	**Slightly Stuck Open**	A slightly stuck-open bypass valve would divert some flow, potentially leading to lower combustion pressure. However, this should also cause a more significant drop in the `Flow_Pump` reading, which is not observed.
5	**System Contamination**	**Minor Debris**	A very small piece of debris could be causing a minor internal leak in a component like an injector seal, leading to the observed pressure loss. This is a plausible underlying cause for the leak itself.

3. Most Probable Hypothesis

The most probable root cause is a **Minor Leak in the Combustion Chamber or at an Injector Nozzle.**

Reasoning: This single physical fault provides the most coherent explanation for the entire set of symptoms. The primary anomaly is the consistent -10% pressure loss in the combustion chamber. A small, physical leak is the most direct cause for such a pressure drop. The control system's reaction—a slight increase in pump and bypass flow—is a logical attempt to compensate for the leak. The fact that the `Flow_Engine` readings remain nominal suggests the control system is successfully maintaining the commanded fuel-to-air ratio despite the leak. This points away from a sensor error and towards a real, but minor, physical fault that the system is actively managing.